Thermal Band Indicators of Crop Heat Stress Before NDVI Decline

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INTRODUCTION

Early detection of drought stress is critical for preserving crop yield. While NDVI reflects vegetation health, it often lags behind biological stress responses. In contrast, canopy temperature increases rapidly under water deficit, making thermal infrared data a potential early indicator (Jackson et al., 1981). This study investigates whether Landsat 8 surface temperature can detect early drought stress in soybean fields in Champaign County, Illinois. Soybean, a major Illinois crop and abundant locally, is especially vulnerable to mid-season heat and moisture limitations. Improving early stress detection can guide precision irrigation and mitigate yield loss in a changing climate.

RESEARCH QUESTION

Can thermal bands detect drought symptoms following abnormal heat for crop health before visible NDVI symptoms appear?

STUDY SITE & DATA

We chose **Champaign County, IL** as our study site to keep the analysis local and because we are familiar with the potential extreme temperatures. The data was collected from USGS Earth Explorer and NOAA weather data. We used the NOAA data to find the most extreme temperature dates from July-August 2015, which was then what we used to choose days for the landsat data. We decided on July-August 2015 as 2015 was a big drought year and July-August is the most crucial period for soybean growth (stages R2-R6). The crop we decided to analyze was soybean as it is quite abundant in Illinois, specifically Champaign County.

TEAM MEMBER CONTRIBUTIONS

Anika Khandavalli
- Weather data acquisition/cleaning
- ROI creation

Ethan Crawford
- Image preprocessing (subsetting/LST calculation)
- Data analysis (regression & tables)

Xylia Greeson
- NDVI calculations
- Literature review

METHODS

Data Acquisition

- Landsat 8 Collection 2 Level-2 Images, 07/24/15 & 08/25/15 (USGS EarthExplorer)
- NOAA Air Temperature and Rainfall Data
- 2015 CropScape Soybean Classification Layer
- Champaign County Shapefile (USCB ACS TIGER Boundaries)

Preprocessing (ENVI 5.3)

- Subset Landsat & classification images to Champaign County
- Layer Stack: B4 + B5 → Compute NDVI
- Convert B10 to LST (℉) using Band Math
- Stack NDVI + LST per each date

ROI Selection

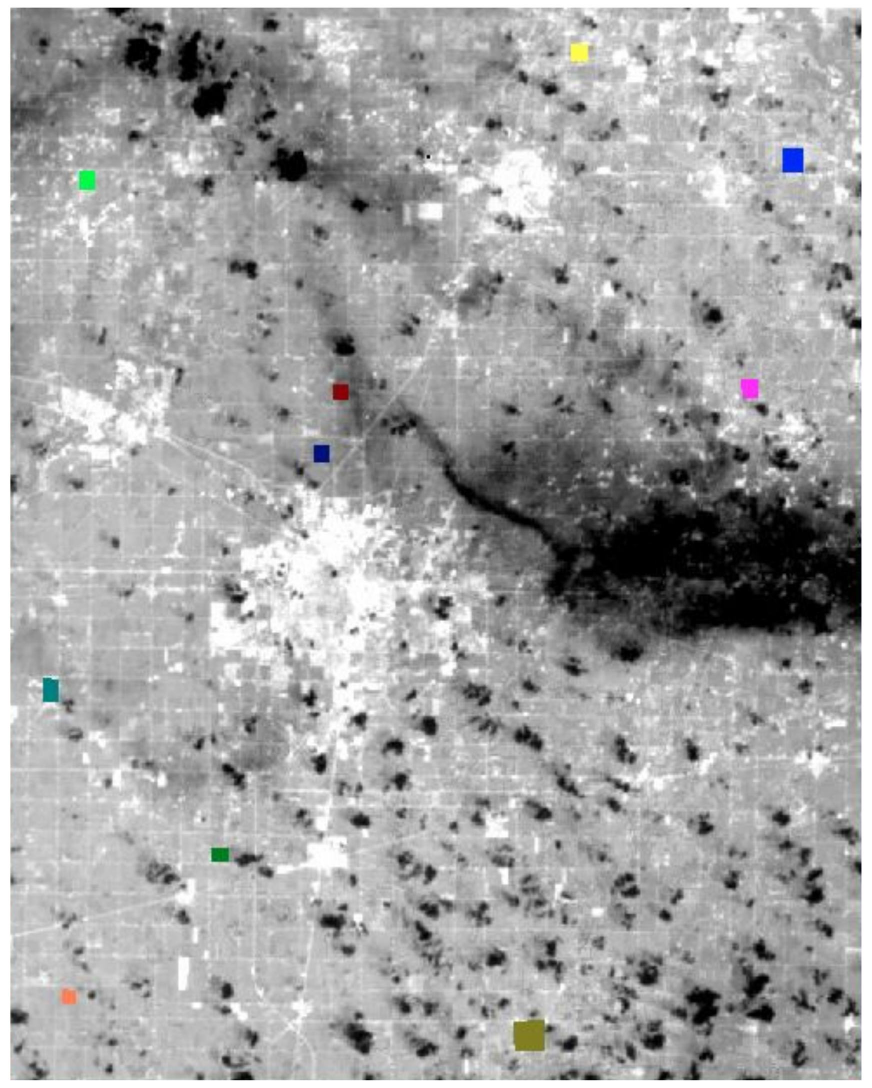
- Link soybean classification layer to Landsat images
- Digitize 10 soybean field ROIs (ENVI ROI tool)
- Save single .roi file for use with both dates

Statistical Extraction (ENVI Classic)

- ROI Statistics for each field with each date
- Min, Mean, Max NDVI & LST values, exported as .txt
- Manually input data into Excel

Analysis (Excel)

- Summarize max air temperature (mean) and rainfall (total) (07/24 - 08/25)
- Find 3-day high temperature over time frame (89℉)
- Compute ΔNDVI and ΔLST per field
- Normalize data for linear regression analysis



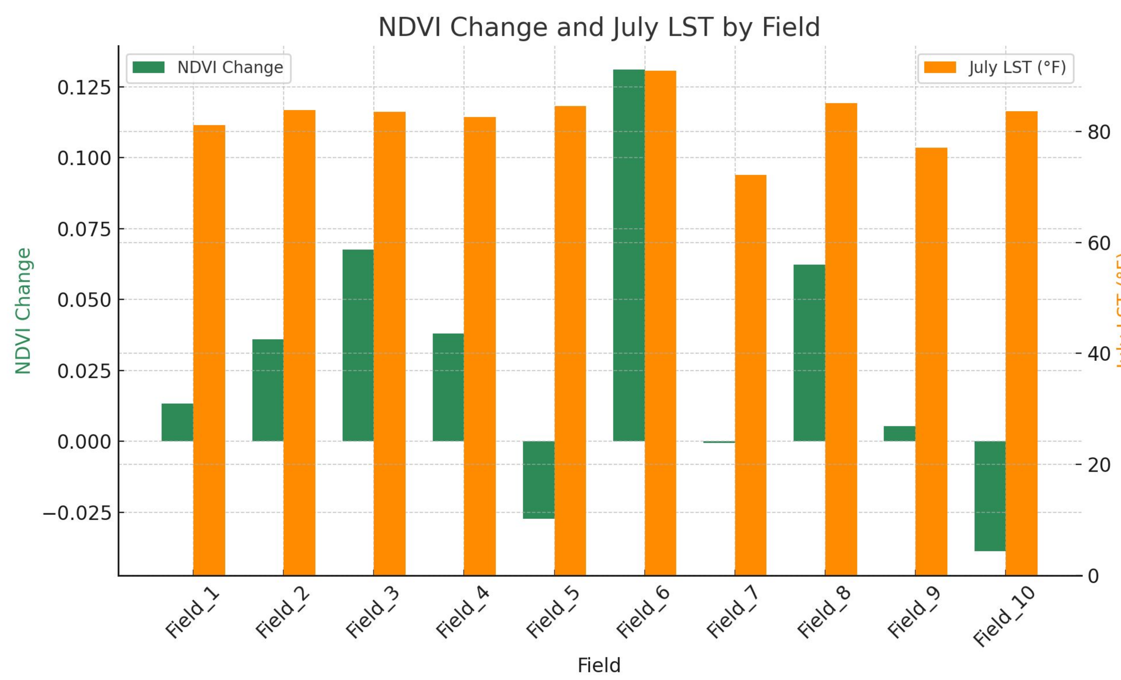
Champaign County Soybean Field ROI Selections

RESULTS

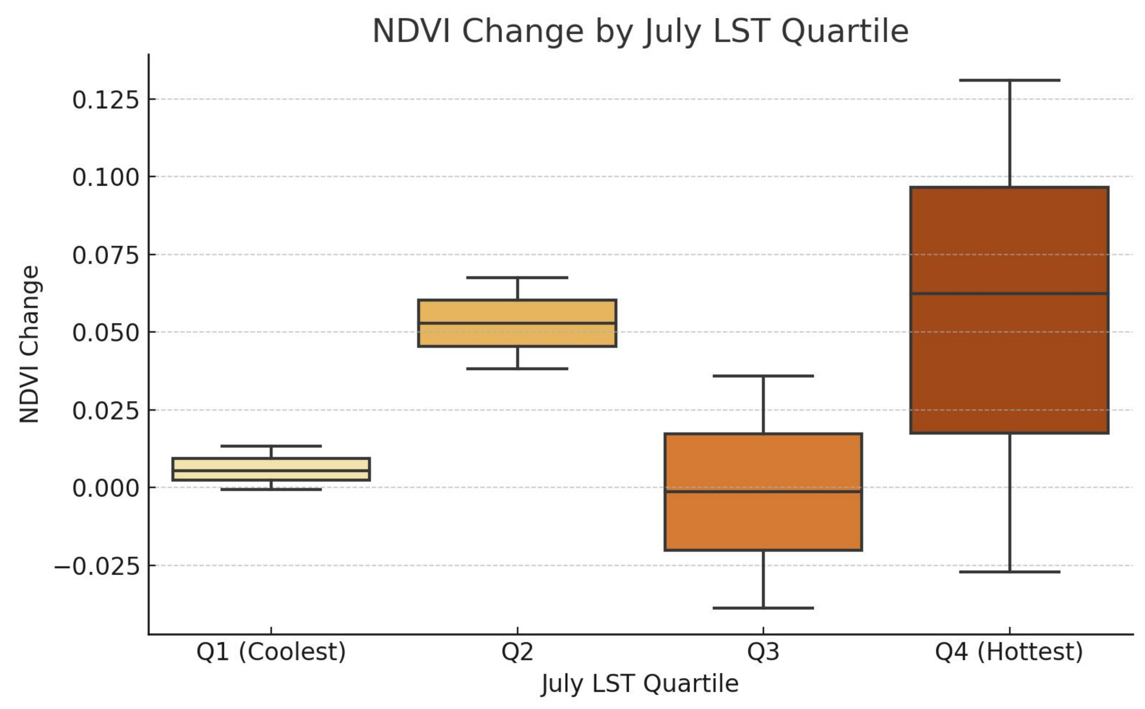
Across the ten soybean fields examined, most showed an increase in NDVI between July 24 and August 25, suggesting continued crop development or possible recovery following early-season heat. While this trend did not reflect classic stress symptoms, variability in the magnitude of NDVI change indicated differential field response.

Regression analysis revealed a modest positive relationship between July LST and NDVI change (R^2 = 0.31, $p \approx 0.09$), suggesting that warmer fields in late July sometimes experienced greater greening by late August—potentially due to lagging growth rather than stress. Fields exposed to lower LST tended to show smaller NDVI gains, though the relationship was not statistically strong. Rainfall and air temperature data, which were uniform across all fields, showed no meaningful predictive power in the model.

Overall, thermal conditions showed limited but detectable influence on field-level NDVI trajectories, highlighting the complexity of interpreting remote sensing signals of crop stress over short periods.



Mean NDVI change (Aug–Jul) and July 24 surface temperature (LST) for each of the ten soybean field ROIs. NDVI change is shown in green; July LST is shown in orange. Fields with higher LST did not consistently show reduced NDVI, suggesting variable crop response to thermal conditions across the county.



Boxplot showing NDVI change grouped by quartiles of July LST. While some fields with higher LST showed greater NDVI gains, the relationship was not consistent or statistically significant, indicating that NDVI increases may reflect growth progression rather than direct drought recovery.

LIMITATIONS

- Only 10 fields were analyzed
- NDVI increased in most fields, limiting detection of stress
- County-wide weather data lacks spatial specificity
- Constant variables (e.g., rainfall) limit regression power

CONCLUSIONS

- Landsat thermal data may indicate early-stage drought stress in soybeans, but more robust NDVI decline patterns are needed for conclusive links.
- Field-based differences in NDVI change suggest local variability in recovery or phenological stage.
- Future work should expand sample size, include multiple years, and integrate higher-resolution weather data.

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